Model Theory

Overview

This section describes how SWAT+ represents the physical and management processes that move water, sediment, nutrients, and other constituents through a watershed. Each page focuses on one process cluster, lists the governing relationships, the basin codes and parameters that switch behaviour, and the source modules in `swatplus/src/` that implement it.

The clusters group together because they share state, run in the same daily order, or are configured through the same input files. The water balance cluster (HRU hydrology, soils, climate, erosion) feeds the routing cluster (channels, aquifers, reservoirs and wetlands, recall objects). The biogeochemistry cluster (plant growth, nutrient cycling, carbon, pesticides, salts and constituents) sits inside the HRU. Decision tables drive management and release rules across all clusters.

Pages in this section

Cluster	Page	What it covers
Water balance	Overview and water balance	Daily HRU water balance, object flow chart, sequencing
Water balance	Climate and weather	Measured, simulated, and lapsed weather; PET methods
Water balance	Hydrology	Runoff (CN, Green-Ampt), lateral flow, percolation, channel routing
Water balance	Soil water and temperature	Soil profile water, soil temperature, frozen soil
Water balance	Erosion and sediment	MUSLE on land, channel erosion and deposition
Routing	Channels	Channel routing methods, sediment transport, in-stream water quality
Routing	Aquifers	1D recession aquifer and 2D `gwflow` cell-based groundwater
Routing	Reservoirs and wetlands	Reservoir water balance, release rules, sediment and nutrient processes; HRU wetlands
Routing	Recall and point sources	Recall time series, export coefficients, delivery ratios
Routing	Decision tables	Runtime condition evaluation and action firing
Biogeochemistry	Plant growth and land use management	Heat units, biomass, yield, scheduled management
Biogeochemistry	Nutrient cycling	Soil N and P pools, transformations, transport
Biogeochemistry	Carbon	Static, CFARM, and Century carbon models
Biogeochemistry	Pesticides	Application, partitioning, decay, transport
Biogeochemistry	Salts and constituents	Salt ion chemistry and generic constituents

How to read these pages

Each page follows the same shape. The Overview names the process and the role it plays in the simulation. Process equations give the governing relationships in plain math. Switches and parameters lists the `codes.bsn` and `parameters.bsn` entries that change the result, with defaults from the source. Implementation lists the source files in `swatplus/src/` that perform the calculation. Related pages cross-link to the input and output references.

The Fortran source at `swatplus/src/` is the source of truth. Where this section conflicts with the source, the source wins. Anything still uncertain is marked `TODO: verify`.